

Comparison of reflected GPS wind speed retrievals with dropsondes in tropical cyclones

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Received 7 June 2009; revised 20 July 2009; accepted 23 July 2009; published 3 September 2009.

[1] In an earlier communication, data were presented that demonstrated that quasi-specular, L-Band reflection measurements could be used to infer ocean surface winds. Applying an indirect calibration technique, a mean square slope versus surface wind speed was developed and reported. Retrievals using this calibration showed that the resulting surface wind speeds were comparable with other measurements. This report extends the previous results by presenting direct comparisons between GPS dropwindsonde (dropsonde)-reported wind speeds and the Bi-static GPS wind speed retrievals for data sets acquired in 2008. Editing of the Bi-static GPS data will be discussed that takes into effect overland and inside-the-eye winnowing. Data will be presented with a regression line to determine the comparative relationship. It will be shown that good agreement exists between the reflected Bi-static GPS retrieved winds and those reported by the dropsondes when certain well-defined types of data are excluded. **Citation:** Katzberg, S. J., and J. Dunion (2009), Comparison of reflected GPS wind speed retrievals with dropsondes in tropical cyclones, *Geophys. Res. Lett.*, **36**, L17602, doi:10.1029/2009GL039512.

1. Introduction and Background

[2] Since 1998 a continuing effort has been pursued to determine the utility of GPS signals reflected from the ocean surface [Garrison *et al.*, 1998; Lin *et al.*, 1999; Katzberg and Garrison, 2000; Katzberg *et al.*, 2001]. A modified GPS receiver was used to acquire reflected data from the ocean surface to provide input to a wind speed retrieval algorithm. The basis for the wind speed retrievals in these previous studies was by means of matching receiver data to a set of model waveforms. The shape of those reference waveforms is controlled via a linear wind speed mean square slope (m.s.s.) relationship originally developed by Cox and Munk [1954]. As suggested by Wilheit [1979] the m.s.s. is adjusted to take into account the fact that the wavelength of the L-Band wavelength (~ 19 cm) is much greater than optical wavelengths. Early results from these experimental studies showed good agreement with buoys and other surface observations, although available experimental conditions were confined by available flight opportunities to winds below 20 m s^{-1} . During and after the 2000 Atlantic hurricane season, it became possible to routinely fly GPS reflection receivers on one or another of NOAA's P-3 Orion research aircraft flying out of

MacDill AFB, Florida. As a result of these flights into tropical cyclones, data have been acquired at high wind speeds ($>20 \text{ m s}^{-1}$).

[3] During the succeeding eight years, a process of continual improvement in software and operational settings was carried out. As a result, data sets of superior quality began to become available after 2002. Examination of the Bi-static GPS surface reflection (Bi-static GPS) wind retrievals from these improved data sets showed an underestimation of high surface winds. Two possibilities were considered: First that the m.s.s. exhibited a saturation effect for winds above 20 m s^{-1} , or second that the (scaled) Cox and Munk linear relationship did not represent the variation of the m.s.s. above 20 m s^{-1} . Evidence for a saturation effect for ocean surface roughness has been reported for Ku- and C-Band [Donnelly *et al.*, 1999] although for the C-Band case the saturation was not complete and was incidence angle dependent.

[4] In order to investigate the underestimation of the Bi-static GPS surface wind speed in high wind conditions, a study was initiated. A calibration effort was conducted by examining data from outbound and inbound legs of NOAA P-3's flying through various tropical cyclones (avoiding data from the near-eye and eye areas.) Wind speeds on such flights typically increase from some low value at the staging point to near hurricane force (32 m s^{-1}) just outside the eye wall. Surface "truth" was taken to be the U.S. Navy's COAMPS model, which provides high enough surface resolution to ensure that the wind fields are at the spatial scale of that captured by the Bi-static GPS wind retrieval technique. The results of this study [Katzberg *et al.*, 2006] showed that a saturating process was beginning to occur above $15\text{--}20 \text{ m s}^{-1}$, but that the dependence of m.s.s. still increases monotonically with wind speed, consistent with findings of Donnelly *et al.* [1999]. A relationship was developed from this data to give a new L-Band variation of m.s.s. with surface wind speed.

[5] Several flights into tropical cyclones have been completed since the 2000 Atlantic hurricane season and the new calibration has been used to generate wind speed retrievals which are qualitatively consistent with other sources of surface winds such as surface adjusted flight-level winds and the Step Frequency Microwave Radiometer [Uhlhorn *et al.*, 2007].

[6] An additional source of ocean surface wind data is the GPS dropwindsonde (dropsonde). The dropsonde is an aircraft expendable instrument deployed from the NOAA P-3's that provides information on pressure, thermodynamics and wind from the aircraft flight-level to the surface at a vertical resolution of $\sim 5 \text{ m}$ [Hock and Franklin, 1999]. Occasionally the dropsonde fails in deployment, does not report data at the surface, or

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otherwise fails. Nevertheless the dropsonde is considered the standard for surface wind determination. Unfortunately, dropsondes are costly, and report winds only along the trajectory they follow. The dropsondes take from a few to several minutes to complete their fall, depending on the aircraft flight-level. Thus the reported wind speed can be significantly different from that determined contemporaneously at or on the surface below the P-3 (e.g., from surface adjusted flight-level winds or Stepped Frequency Microwave Radiometer data.).

2. Comparison of Reflected GPS Wind Speed Retrievals With Dropsonde Values

2.1. GPS Data and Considerations for Selection

[7] The retrieval of wind speed from the Bi-static GPS is dependent on the increasing mean square slopes induced by surface winds. For example, in the presence of local coastline or islands, the surface roughness can be affected. Three classes of expected deviation from the dropsonde data are noted and were not included in the Bi-static GPS-dropsonde comparison dataset:

[8] 1. In the lee of a landmass, the Bi-static GPS retrieved winds will show an expected decrease in retrieved wind speed the closer the measurement is made to the land. Dropsondes would generally show a similar, but not necessarily identical effect.

[9] 2. In the eye of a tropical cyclone, the surface roughness does not drop precipitously to the value associated with the low winds well away from the storm, since the local ocean is still being stimulated by waves propagating from the nearby eye wall. Dropsondes would record the true low wind speeds.

[10] 3. The Bi-static GPS wind retrieval technique, cannot, for obvious reasons, monitor winds over land, while flight-level and dropsonde winds can still be accurately recorded.

2.2. Dropsondes

[11] Dropsondes [Hock and Franklin, 1999] have been deployed from NOAA P-3 Orion research aircraft since 1997. Although these instruments typically report a surface wind (10 m), these single point measurements represent a semi-Lagrangian instantaneous wind value that may not well-represent the maximum 1-min sustained 10-m wind that is utilized by the NOAA National Hurricane Center (NHC). Therefore, NHC has also adopted the use of the WL150 wind, a low-layer mean dropsonde wind measurement developed by Franklin *et al.* [2003]. This value is derived from the mean of the lowest available 150-m of winds in the dropsonde profile and is then adjusted to the surface using a dropsonde-based mean eye wall profile [Franklin *et al.*, 2003] given by:

$$r(z) = 1.0314 - 0.004071z + 2.465 \cdot 10^{-5}z^2 - 5.446 \cdot 10^{-8}z^3$$

where $r(z)$ is the ratio of the dropsonde 10-m surface wind speed to the WL150 wind speed and z is the mean altitude of the 150-m layer (constrained to values of ≤ 200 m). The WL150 wind and mean altitude of the

150-m layer (z) were both obtained from the transmitted dropsonde messages.

3. Missions and Data

[12] During the 2008 hurricane season there were sixteen named storms in the Atlantic Basin. Of these, Bi-Static GPS data were acquired from five hurricanes or tropical storms and included one to several penetrations of the individual storms. Generally the data were of good quality with sufficient satellites being tracked in the GPS reflection receiver to provide the position solution required for proper tracking of the reflected signal. Occasionally the host aircraft makes extreme banking turns leading to a loss of the minimum four satellites required for proper receiver operation, but this occurred infrequently during NOAA's 2008 aircraft missions.

[13] From the data collected during the 2008 missions, wind speed retrievals were performed using the approach discussed by Katzberg and Garrison [2000] using the modified mean square slope relationship of Katzberg *et al.* [2006]. Dropsonde data were acquired from the NOAA Hurricane Research Division (HRD). Flight-level data were available for all the flights, although no final wind speed product was available from the HRD website. Surface adjusted flight-level and Step Frequency Microwave Radiometer data are typically used as comparison data sources and flight-level data are useful for determining when the aircraft has entered the eye of the storm (Item 2 mentioned above) as part of the winnowing process.

4. Selections

[14] All flights with Bi-static GPS wind data were processed for surface wind speed using the approach described above. Smoothing of the data were done during the retrieval process for as much as ten seconds but the integration time was not found to have a significant enough effect on the comparisons to warrant separating out for individual consideration. All data presented here is for 5–10 second averaging. Each flight with dropsonde data was identified to build a matched pair of data sets having both Bi-static GPS and dropsonde measurements. Any mission in which a considerable part of the flight was over land, e.g., Gustav, September 1, 2008, was discarded since the GPS Bi-static data would be influenced by un-modeled effects of fetch or over-land non-retrievals.

[15] Dropsondes were evaluated for proper operation, typically meaning that they provided wind speeds all the way to the ocean surface. Three “surface” wind speeds are available: The last value reported, the WL150 value extrapolated to the surface, and a value extrapolated by the National Hurricane Center from the dropsondes to give surface wind speed. The WL150 dropsonde data were processed to yield a surface (10 m) value. If that was not possible, that particular dropsonde was eliminated from the dataset. Occasionally a dropsonde would malfunction for one or another reason (e.g., report a wind speed that showed no altitude profile, or some other anomaly) and was similarly eliminated. For this paper, the WL150 surface extrapolated wind and the NHC dropsonde surface wind (Sfc extrapolated, on the plots) were used.

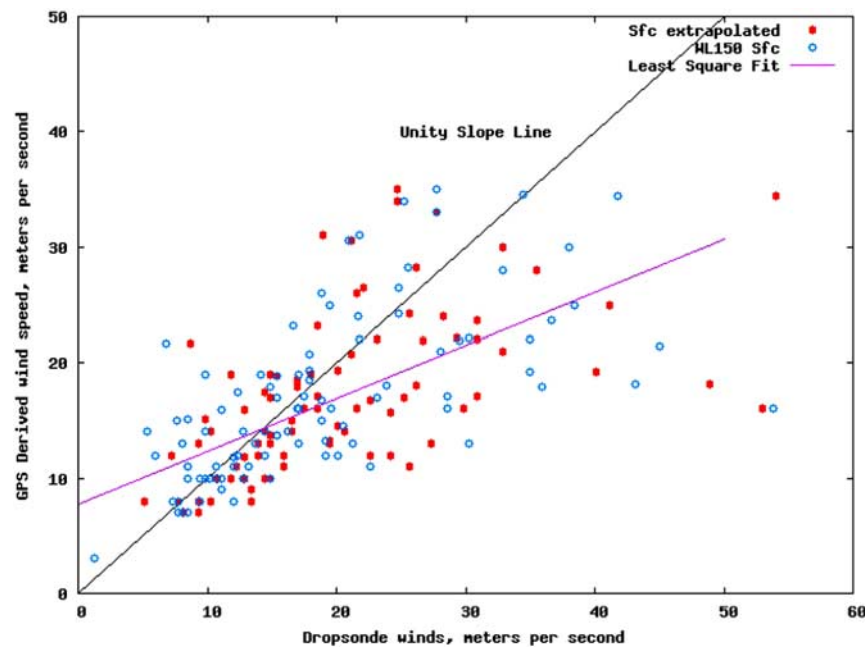


Figure 1. GPS retrievals (y-axis) plotted against dropsonde reported winds (x-axis.) All dropsondes plotted except those found to be over land or in the storm eye. Closed circles are NHC surface extrapolated (Sfc extrapolated), open circles WL150 surface estimate. Linear least squares fit line with slope of 0.46 and y-intercept 7.7 ms^{-1} . Standard deviation around the fit is 5.8 ms^{-1} .

[16] Data from 117 dropsondes were collected that had a GPS surface reflection match: 14 from Fay August 14th; 13 from Fay August 19th; 14 from Hurricane Gustav August 31st; 14 from Ike September 10th; 17 from Ike September 17th; 5 from Tropical Storm Kyle September 6th; 7 from Kyle, September 27th; 23 from Paloma November 7th and 10 from Paloma November 8th.

5. Results

[17] Using the editing process discussed above a comparison was made between the GPS-derived values and the dropsonde data. Figure 1 represents the result of removing dropsonde-Bi-static GPS retrieval pairs over land or inside the tropical cyclone eye. The total number of Bi-static GPS-to-dropsonde comparisons after editing was 110. As mentioned before, these pairs were determined by identifying near zero values of flight-level winds and an eye-wall. Although the winds are relatively calm in the eye, the surface roughness is enhanced in this part of the storm due to propagation of waves from the eye wall. Therefore, the Bi-static GPS retrievals will tend to overestimate the wind in the region of the eye. A linear-least-squares fit line with slope of 0.46 and y-intercept 7.7 ms^{-1} is shown in Figure 1. Standard deviation around the fit is 5.8 ms^{-1} .

[18] One of the characteristics of the Bi-static GPS retrieval technique is the fact that wind speed retrievals will not infrequently yield their peak winds at a location somewhat different from those of flight-level winds or SFMR. The reason that the peak surface roughness does not necessarily occur at the same location as peak surface winds is yet to be determined. Occasionally a storm will be found that yields abnormally low GPS-derived peak winds. Gustav and Paloma were two such storms. The result of

excluding these two storms is shown in Figure 2. A best fit to the data without Gustav and Paloma is shown in Figure 2. All dropsondes are plotted except Gustav, August 31, 2008 and Paloma, November 7th and 8th. Closed circles are NHC surface extrapolated (Sfc extrapolated), open circles WL150 surface estimate. A least squares fit was also done to this data and yielded a slope of 0.83 and a y-intercept of 2.9 ms^{-1} . The mean square deviation from the fit curve was 4.8 ms^{-1} . The difference in number of points is a result of different criteria applied to selecting either WL150 surface extrapolated versus NHC direct surface extrapolation.

6. Discussion and Conclusions

[19] When comparing Bi-static GPS reflection retrieved surface wind speeds with dropsonde data in tropical cyclone environments it is obvious that editing out over-land areas must be done since this technique cannot make measurements there. The same would be true of microwave radiometers or scatterometers since both of these techniques operate via RF interaction with the ocean surface. The existence of surface roughness residual in the eye of tropical cyclones is another of the anomalies found when using the Bi-static GPS technique to retrieve wind speeds in these storms. Removing these anomalous conditions from the Bi-static GPS reflection retrieved wind speeds and dropsonde set yields an improved comparison.

[20] The cause of poor results from the Hurricanes Gustav and Paloma datasets is not known. According to information from the National Hurricane Center, both Gustav and Paloma were undergoing significant changes, which could cause some sort of latency between surface winds and surface roughness development. In the case of Gustav, the storm had just re-emerged into the Gulf of

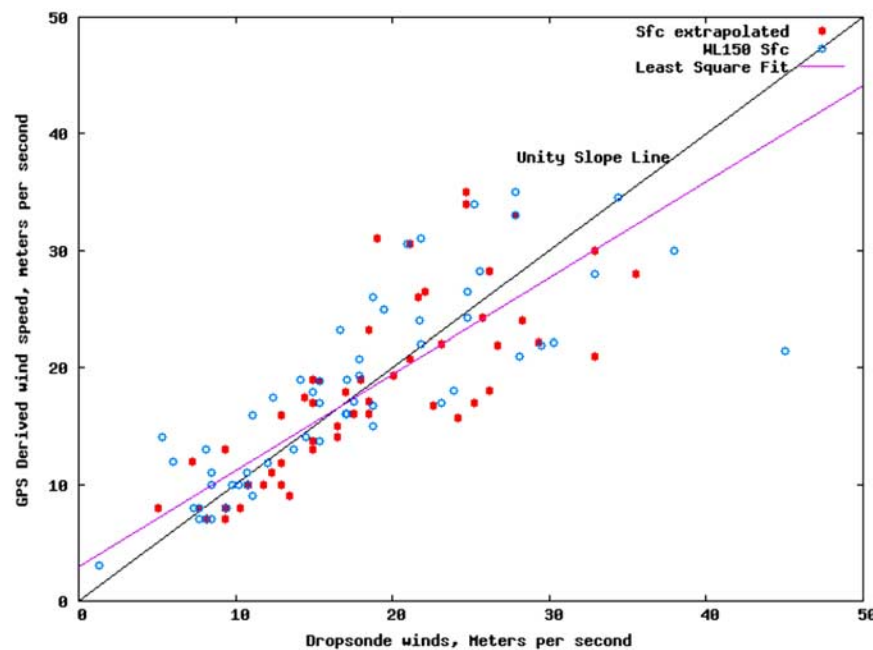


Figure 2. GPS retrievals (y-axis) plotted against dropsonde reported winds (x-axis.) All dropsondes plotted except Gustav, August 31, 2008 and Paloma, November 7th and 8th. Closed circles are NHC surface extrapolated (Sfc extrapolated), open circles WL150 surface estimate. The slope for the least squares line is 0.83 with a y-intercept of 2.89 ms^{-1} . The standard deviation around the fit is 4.8 ms^{-1} .

Mexico after traversing Cuba. Rather than intensify, the storm underwent a weakening and expansion. From the National Hurricane Center Tropical Cyclone Report “It (Gustav) rapidly intensified to a Category 4 hurricane before it made landfall on the eastern coast of the Isle of Youth, Cuba, near 1800 UTC that day (August 30). . . and Gustav weakened over Cuba, and it continued to weaken over the Gulf of Mexico on 31 August.”

[21] For Hurricane Paloma the NHC Tropical Cyclone Report stated “During the 24-h period ending at 1200 UTC 8 November, Paloma’s intensity increased by 50 kt” and “During the 24-h period ending at 1200 UTC 9 November, Paloma weakened by 90 kt, from a 125-kt category 4 hurricane (Figure 6) to a 35-kt tropical storm. . .”

[22] None of the other storms reported here were subject to equivalent major changes in such a short time. In addition, one of the two runs into Gustav showed good agreement with flight-level winds and SFMR except in one of the three legs of that one flight. Unfortunately the nine dropsondes currently available from HRD are from the wrong P-3. The other flight which was deleted (in which Gustav was weakening) had generally poor agreement but as many as 18 dropsondes were recorded, accentuating the disagreement. Thus, we felt justified in showing both the with Paloma-Gustav and the without comparisons.

[23] The Bi-static GPS retrievals are dependent on sampling areas in which the m.s.s. has reached a level indicative of the wind speeds in the storm. Discrepancies between the two wind speeds might arise from the surface reflection spot missing the areas of ultimate m.s.s. or dropsonde splash location too different from the GPS surface reflection point. This would especially be an issue for dropsondes launched in the hurricane eye wall, where these instruments can advect 15–20 km or more from their flight-level launch point to

where they splash at the ocean surface. Currently the occasional poor match of wind speeds is a subject of investigation with as yet no satisfactory explanation.

[24] The advection of the dropsondes by the winds causes their final approach to the surface to be separated from the Bi-static GPS reflection point. Nevertheless, agreement between the GPS and dropsondes is generally good and confirms the quantitative nature of the modified Cox and Munk calibration developed previously. Correcting for the separation or at least setting tolerances for accepting a dropsonde-Bi-static GPS reflection retrieval separation have not been done and will be left for later investigation.

[25] Also of interest is the preponderance of the GPS wind speeds below the unity slope line in Figure 1. The calibration that was done using the COAMPS approach resulted in a model curve that appears to still underestimate the higher winds. The deviation from the unity line is not severe and suggests that correction of the m.s.s. versus wind speed using data taken by dropsondes rather than using models could improve retrievals.

[26] This comparison of dropsondes with GPS reflection derived wind speeds has shown that the GPS technique gives good agreement with surface “truth” with r.m.s errors on the order of 5 ms^{-1} for winds ranging to 40 ms^{-1} . This level of accuracy is attainable if certain clearly identifiable cases are avoided such as within the eye and over land. In addition, correcting effects of advection in displacing the dropsondes spatially from the GPS reflection point could yield improved comparisons. Effects of fetch are not yet taken into account and would be expected to adversely affect wind speed comparisons. Finally, certain storms occasionally do not yield the strong wind speeds found by other methods. The cause of this is a subject left for investigation.

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